

This question paper contains 4 printed pages]

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S. No. of Question Paper : 2251

Unique Paper Code : 32351402

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Name of the Paper : Riemann Integration and Series of Functions

Name of the Course : B.Sc. (H) Mathematics

Semester : IV

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt any two parts from each question.

All questions are compulsory.

1. (a) Find the upper and the lower Darboux integrals for $f(x) = 2x^3$ on $[0, 1]$. Is f integrable on $[0, 1]$? Justify. 6
- (b) Prove that a bounded function f on $[a, b]$ is Riemann integrable if and only if it is Darboux Integrable and the value of integrals agree. 6
- (c) Prove that if f is integrable on $[a, b]$ then $|f|$ is integrable on $[a, b]$ and $\left| \int_a^b f \right| \leq \int_a^b |f|$. What about the converse? Justify your answer. 6
2. (a) Prove that every bounded piecewise monotonic function f on $[a, b]$ is integrable. 6.5

P.T.O.

- (b) State Fundamental Theorem of Calculus II. Let f be defined as follows :

$$f(t) = \begin{cases} 0 & \text{for } t < 0 \\ t & \text{for } 0 \leq t \leq 1 \\ 4 & \text{for } t > 1 \end{cases}$$

Determine the function $F(x) = \int_0^x f(t)dt$. Discuss the continuity and differentiability of F and also calculate F' . 6.5

- (c) Let f be a bounded function on $[a, b]$. If P and Q are partitions of $[a, b]$ such that $P \subseteq Q$, show that :

$$L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P). \quad 6.5$$

3. (a) Examine the convergence of the following improper integrals :

$$(i) \int_1^{\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx$$

$$(ii) \int_{-1}^1 \frac{dx}{x^3}.$$

6

- (b) Prove that :

$$\int_0^{\infty} \frac{\sin x}{x} dx$$

is convergent.

6

- (c) Examine the convergence of the improper integral :

$$\int_0^1 x^{a-1} (1-x)^{b-1} dx.$$

6

4. (a) Let (f_n) be a sequence of functions on $[a, b]$ converging uniformly to f on $[a, b]$. Prove that f is integrable on $[a, b]$ and

$$\int_a^b f = \lim \int_a^b f_n. \quad 6.5$$

(b) Let $f_n(x) = \frac{nx}{1+n^2x^2} \quad \forall x \in \mathbb{R}.$

(i) Show that (f_n) converges pointwise to $f(x) = 0$ on $\mathbb{R}.$

(ii) Does (f_n) converge uniformly to f on $[0, 1]$? Justify.

(iii) Prove that (f_n) converges uniformly to f on $[1, \infty[.$ 6.5

(c) Let $f_n: [0, 1] \rightarrow \mathbb{R}$ be defined for $n \geq 2$ by

$$f_n(x) = \begin{cases} n^2x & \text{for } 0 \leq x \leq 1/n \\ -n^2(x - 2/n) & \text{for } 1/n \leq x \leq 2/n \\ 0 & \text{for } 2/n \leq x \leq 1. \end{cases}$$

Show that the sequence (f_n) is not uniformly convergent. 6.5

5. (a) Let (f_n) be a sequence of continuous functions on a set $A \subseteq \mathbb{R}$. If (f_n) converges uniformly to f on A , prove that f is continuous on A . 6

- (b) Prove that the series :

$$\sum_{n=1}^{\infty} \sin\left(\frac{x}{n^2}\right)$$

is uniformly convergent on $[-a, a]$, $\forall a > 0$, but is not uniformly convergent on $\mathbf{R}$. 6

- (c) Prove that the series :

$$\sum \frac{\cos(x^2 + 1)}{n^3}$$

represents a continuous function on $\mathbf{R}$. 6

6. (a) Find the radius of convergence of the power series :

(i) $\sum_{n=1}^{\infty} \frac{2^n}{n^2} x^n$

(ii) $\sum_{n=1}^{\infty} x^{n!}$. 6.5

- (b) Let $\sum a_n x^n$ has radius of convergence $R > 0$ and let :

$$f(x) = \sum a_n x^n \text{ for } |x| < R.$$

Then prove that f is differentiable on $(-R, R)$ and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} \text{ for } |x| < R. \quad 6.5$$

- (c) Prove that :

$$\sum_{n=1}^{\infty} n^2 x^n = \frac{x(x+1)}{(1-x)^3}$$

for $|x| < 1$. Hence evaluate :

$$\sum_{n=1}^{\infty} \frac{n^2}{3^n}. \quad 6.5$$